ClipSAT : ACT Subject Test Math 1 [ANSWER KEY]

Unit 0.01: ACT II Math Level 1 Diagnostic & Strategy Drill

Q1. In the standard x, y coordinate plane, the lines with equations 2x - y = 7 and x + y = 5 intersect at the point a, b. What is the value of a?

[ ] A: 1
[ ] B: 3
[x] C: 4
[ ] D: 5

Solution: The point of intersection a, b satisfies both linear equations simultaneously: 1 2a - b = 7 2 a + b = 5 To solve for a, add the two equations together to eliminate b: 2a - b + a + b = 7 + 5 3a = 12 a = 4 Substituting a = 4 into the second equation gives 4 + b = 5, which yields b = 1. Thus, the lines intersect at 4, 1, and the value of a is 4. - A common misconception is solving for b = 1 and selecting 1 Choice A. - Another error is an arithmetic mistake in dividing 12 by 3 to get 3 Choice B. - Choice D represents the constant from the second equation without solving the system.

Q2. In right triangle ABC, altitude overline{BD} is drawn to hypotenuse overline{AC}. If AD = 4 and DC = 9, what is the length of altitude overline{BD}?

[x] A: 6
[ ] B: 6.5
[ ] C: 13
[ ] D: 36

Solution: By the Geometric Mean Theorem Altitude Rule for right triangles, the altitude to the hypotenuse is the geometric mean of the two segments into which it divides the hypotenuse: BD^2 = AD cdot DC Substitute the given lengths AD = 4 and DC = 9: BD^2 = 4 cdot 9 = 36 Taking the positive square root: BD = sqrt{36} = 6 Common Misconceptions: - Choice B 6.5 confuses the geometric mean with the arithmetic mean: frac{4 + 9}{2} = 6.5. - Choice C 13 incorrectly adds the two segments together: 4 + 9 = 13. - Choice D 36 finds BD^2 = 4 times 9 but forgets to take the square root.

Q3. The bar graph below shows the number of students enrolled in each of four elective courses at a high school. If one student is selected at random from these four courses, what is the probability that the selected student is enrolled in Music?

[ ] A: dfrac{7}{50}
[x] B: dfrac{7}{20}
[ ] C: dfrac{7}{13}
[ ] D: dfrac{13}{20}

Solution: To find the probability that a randomly chosen student is enrolled in Music, divide the number of students enrolled in Music by the total number of students enrolled across all four courses. 1. Find the total number of students: text{Total} = 8 + 14 + 6 + 12 = 40 2. Find the number of students in Music: ntext{Music} = 14 3. Calculate the probability: Ptext{Music} = frac{14}{40} = frac{7}{20} Common misconceptions: - dfrac{7}{13} computes the odds in favor ratio of students in Music to students not in Music, frac{14}{26} rather than probability. - dfrac{13}{20} represents the complement probability the probability that the student is NOT enrolled in Music, 1 - frac{7}{20} = frac{13}{20} . - dfrac{7}{50} incorrectly assumes the total is 100 students.

Q4. The graph of the quadratic function fx = x^2 - 6x + 5 is shown in the standard x, y coordinate plane below. A horizontal line y = k intersects the graph of fx at two points that are 8 units apart. What is the value of k ?

[x] A: 12

[ ] B: 8

[ ] C: 4

[ ] D: -3

Solution: We are given the quadratic function fx = x^2 - 6x + 5 and a horizontal line y = k that intersects the parabola at two points separated by a horizontal distance of 8 . **Method 1: Using the axis of symmetry** The axis of symmetry for a parabola in standard form y = ax^2 + bx + c is given by: x = -frac{b}{2a} = -frac{-6}{21} = 3 Because parabolas are symmetric about their axis of symmetry, the two intersection points on the horizontal line y = k must lie equidistant from x = 3 . Since the total distance between them is 8 , each point lies frac{8}{2} = 4 units away from the line of symmetry: x_1 = 3 - 4 = -1 x_2 = 3 + 4 = 7 Since both points lie on the graph of fx , the value of k is found by evaluating fx at either x = -1 or x = 7 : k = f7 = 7^2 - 67 + 5 = 49 - 42 + 5 = 12 k = f-1 = -1^2 - 6-1 + 5 = 1 + 6 + 5 = 12 **Method 2: Using the quadratic formula** Setting fx = k gives the quadratic equation x^2 - 6x + 5 - k = 0 . Using the quadratic formula, the solutions are: x = frac{6 pm sqrt{-6^2 - 415 - k}}{2} = 3 pm sqrt{9 - 5 - k} = 3 pm sqrt{4 + k} The distance between these two roots is the difference: 3 + sqrt{4 + k} - 3 - sqrt{4 + k} = 2sqrt{4 + k} Setting this distance equal to 8 : 2sqrt{4 + k} = 8 implies sqrt{4 + k} = 4 implies 4 + k = 16 implies k = 12 Common misconceptions: - Choosing 8 confuses the horizontal distance between the points with the vertical position k . - Choosing 4 results from solving sqrt{4 + k} = 4 by incorrectly writing 4 + k = 8 instead of squaring 4 . - Choosing -3 results from substituting the half-distance x = 4 directly into fx instead of shifting from the axis of symmetry x = 3 .

Q5. In the standard x, y coordinate plane, the lines with equations 2x - y = 7 and x + y = 5 intersect at the point a, b. What is the value of a?

[ ] A: 1

[ ] B: 3

[x] C: 4

[ ] D: 5

Solution: The point of intersection a, b satisfies both linear equations simultaneously: 1 2a - b = 7 2 a + b = 5 To solve for a, add the two equations together to eliminate b: 2a - b + a + b = 7 + 5 3a = 12 a = 4 Substituting a = 4 into the second equation gives 4 + b = 5, which yields b = 1. Thus, the lines intersect at 4, 1, and the value of a is 4. - A common misconception is solving for b = 1 and selecting 1 Choice A. - Another error is an arithmetic mistake in dividing 12 by 3 to get 3 Choice B. - Choice D represents the constant from the second equation without solving the system.

Q6. In the standard x, y coordinate plane, a circle has its center at -3, 5 and passes through the point 1, 8. What is the radius of this circle?

[x] A: 5

[ ] B: 7

[ ] C: sqrt{13}

[ ] D: 25

Solution: The radius r of a circle is the distance from its center to any point on the circle. Using the distance formula between the center -3, 5 and the point on the circle 1, 8: r = sqrt{x_2 - x_1^2 + y_2 - y_1^2} r = sqrt{1 - -3^2 + 8 - 5^2} = sqrt{4^2 + 3^2} = sqrt{16 + 9} = sqrt{25} = 5 Thus, the radius of the circle is 5. Common misconceptions: - Choice B 7 mistakenly adds the coordinate differences directly: 4 + 3 = 7. - Choice C sqrt{13} makes a sign error in subtracting the x-coordinates, computing 1 - 3^2 + 8 - 5^2 = 4 + 9 = 13. - Choice D 25 confuses the radius r with r^2.

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Q7. If fx = 2x - 1 and gx = x^2 + 3 , what is the value of gf2 ?

[ ] A: 9
[x] B: 12
[ ] C: 13
[ ] D: 21

Solution: To find the composite value gf2 , evaluate from the inside out: 1. Evaluate the inner function at x = 2 : f2 = 22 - 1 = 4 - 1 = 3 2. Substitute this result into the outer function gx : g3 = 3^2 + 3 = 9 + 3 = 12 Common Misconceptions: - 13 results from reversing the order of operations to evaluate fg2 : g2 = 2^2 + 3 = 7 , and f7 = 27 - 1 = 13 . - 21 results from multiplying the two values instead of composing them: f2 cdot g2 = 3 cdot 7 = 21 . - 9 results from squaring f2 but forgetting the constant term +3 in gx .

Q8. What is the smallest positive integer n such that 126n is the square of an integer?

[ ] A: 7
[x] B: 14
[ ] C: 21
[ ] D: 42

Solution: To find the smallest positive integer n such that 126n is a perfect square, first find the prime factorization of 126: 126 = 2 times 63 = 2 times 3^2 times 7 For an integer to be a perfect square, every prime factor in its prime factorization must have an even exponent. In the prime factorization of 126: - The factor 2 has an exponent of 1 odd. - The factor 3 has an exponent of 2 even. - The factor 7 has an exponent of 1 odd. To make all exponents even with the smallest positive integer n , n must provide at least one factor of 2 and one factor of 7: n = 2^1 times 7^1 = 14 Then 126 times 14 = 1764 = 42^2 , which is a perfect square. Common misconceptions: - Choosing 7 only pairs the factor 7, leaving 2 with an odd exponent 126 times 7 = 882 , not a square. - Choosing 21 multiplies by 3 times 7 , leaving 2 unpaired. - Choosing 42 gives 126 times 42 = 5292 , which is not a square and is greater than 14.